

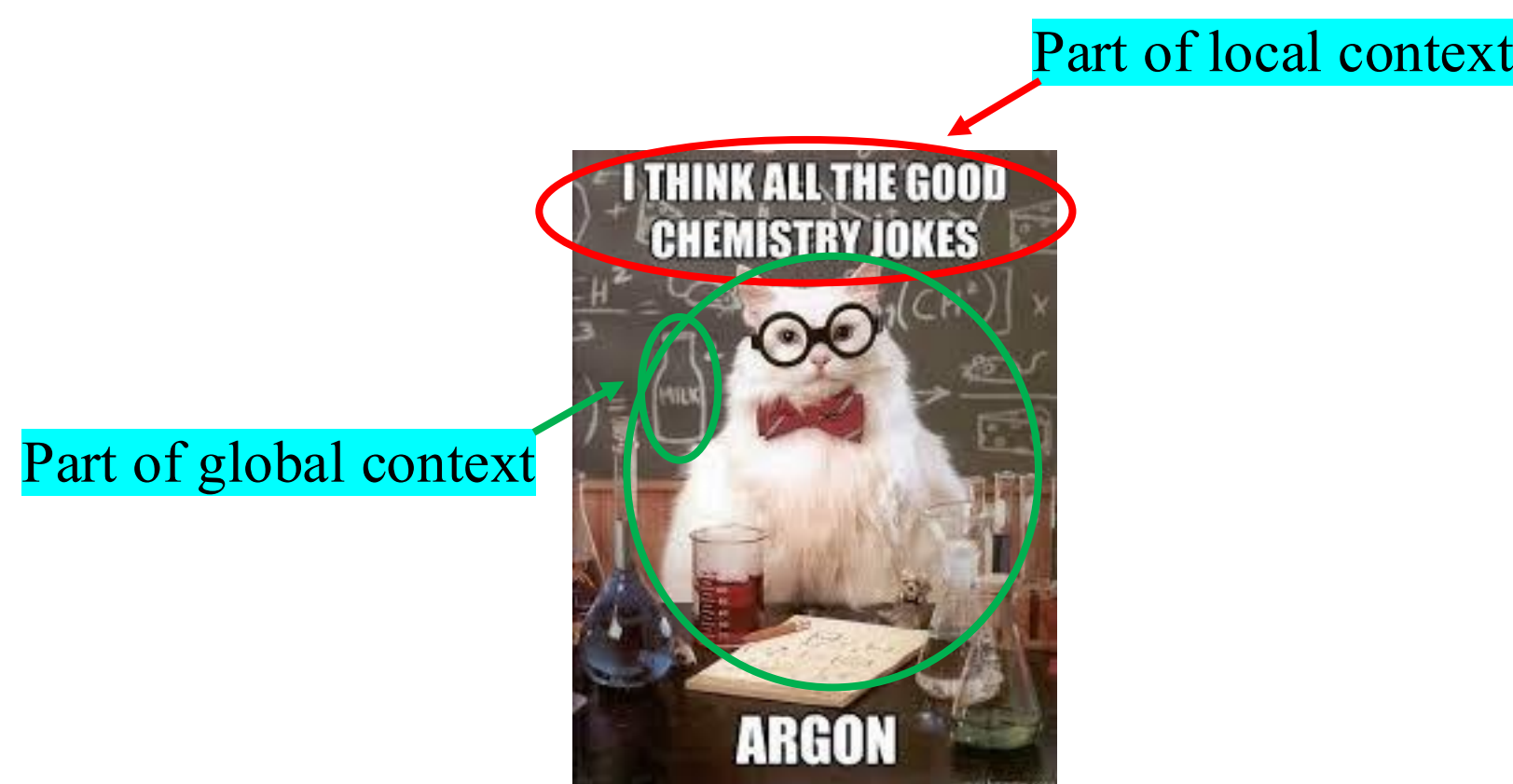
MemeMatch: A Large-Scale Dual-Context Multimodal Dataset and Retrieval System for Internet Memes

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Introduction

MemeMatch is a large-scale dual-context meme dataset + retrieval system that separates **local user text** from **global template semantics**, adds rich annotations (emotion, topic, usage intent), and enables **context-aware search** from text or images.



1. **Local context** = what the meme says (captures the user-added overlay text and the situational message)
2. **Global context** = what the meme shows (captures the underlying image/template semantics without the added text)

Problem

Meme meaning comes from **both** the **template image** and the **user-added text**, so the same template can convey different messages, making search and analysis unreliable with a single representation.

Contributions

1. **Large-scale dataset**: ~301K memes and 2,083 templates collected from **Reddit** (organic social memes) and **ImgFlip** (template-based memes).
2. **Rich annotations**: **14D sentiment/emotion** (11 emotions + 3 polarities), **BERTopic** (300 local topics, 200 global topics), and **28 usage-intent labels**.
3. **Dual-context design**: separates **Local** (overlay text + title) vs **Global** (template caption) to preserve both **message** and **template meaning**.
4. **Context-aware retrieval system**: supports **text + image queries** using **case-based embeddings** and an LLM-based query parser.
5. **Large-scale EDA findings**: reveal trends in **emotion, topics, and usage** over time, and their association with **engagement/virality patterns**.

Dataset Structure

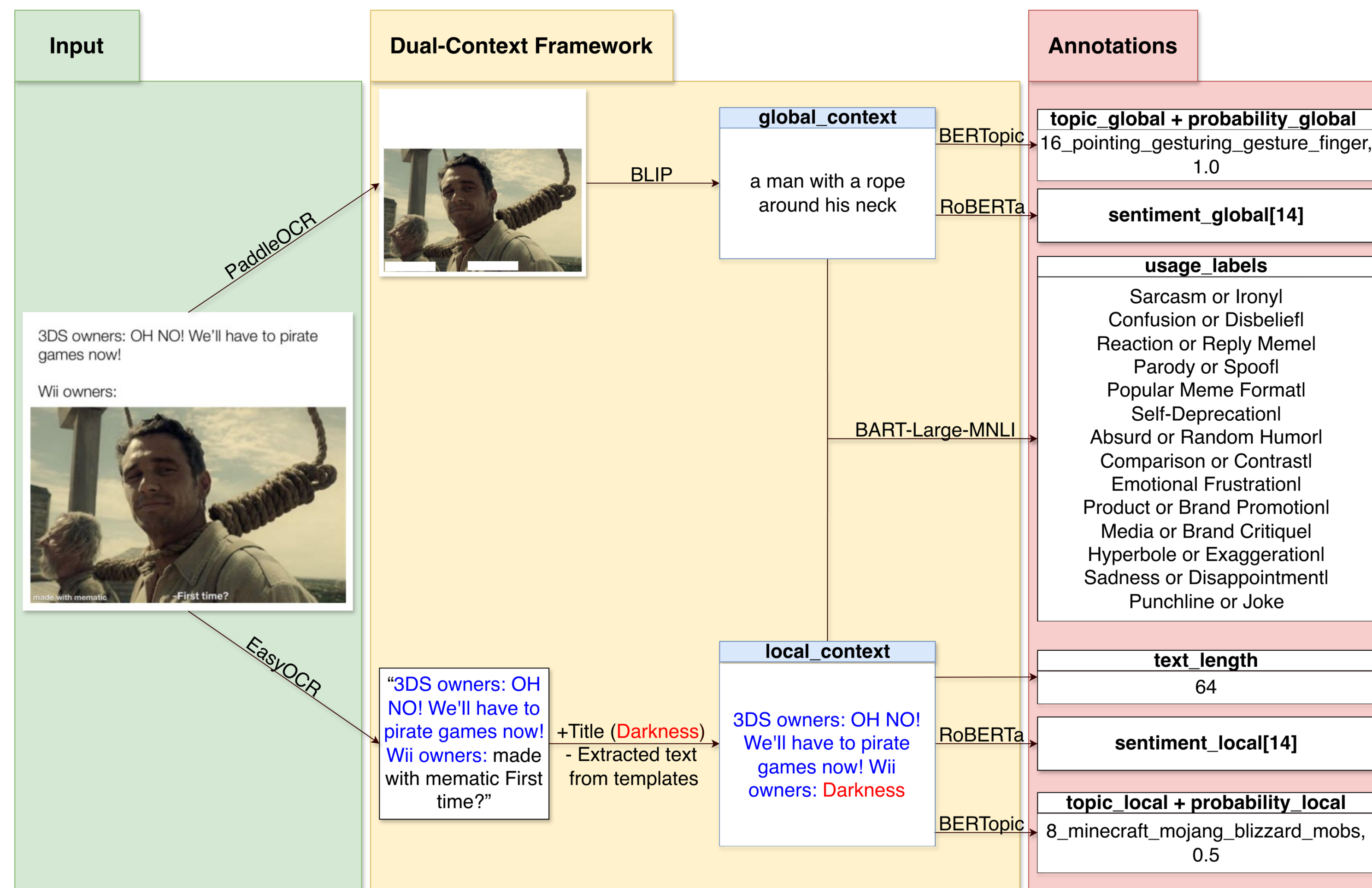
Field	Type	Req.	Description
filename	string	✓	Image filename, used as the unique identifier across all tables. For ImgFlip memes, the filename also indicates the template name.
created_utc	string		Reddit meme upload timestamp (ISO-8601 format).
score	int		Reddit upvote count at crawl time.

Schema (core metadata) for MemeMatch v1.0.

Field	Type	Req.	Description
local_context	string	✓	Cleaned OCR overlay text concatenated with title.
global_context	string	✓	BLIP caption on masked image (template semantics).
text_length	int	✓	Number of characters in local_text.
sentiment_local[14]	float[0..1]	✓	11 emotions + 3 polarities for local text.
sentiment_global[14]	float[0..1]	✓	11 emotions + 3 polarities for global text.
topic_local	string		Topic label assigned by BERTopic for the local text.
topic_global	string		Topic label assigned by BERTopic for the global caption.
topic_score_local	float[0..1]	✓	Topic confidence score for the local context.
topic_score_global	float[0..1]	✓	Topic confidence score for the global context.
usage_labels	string	✓	Zero-shot usage tags.

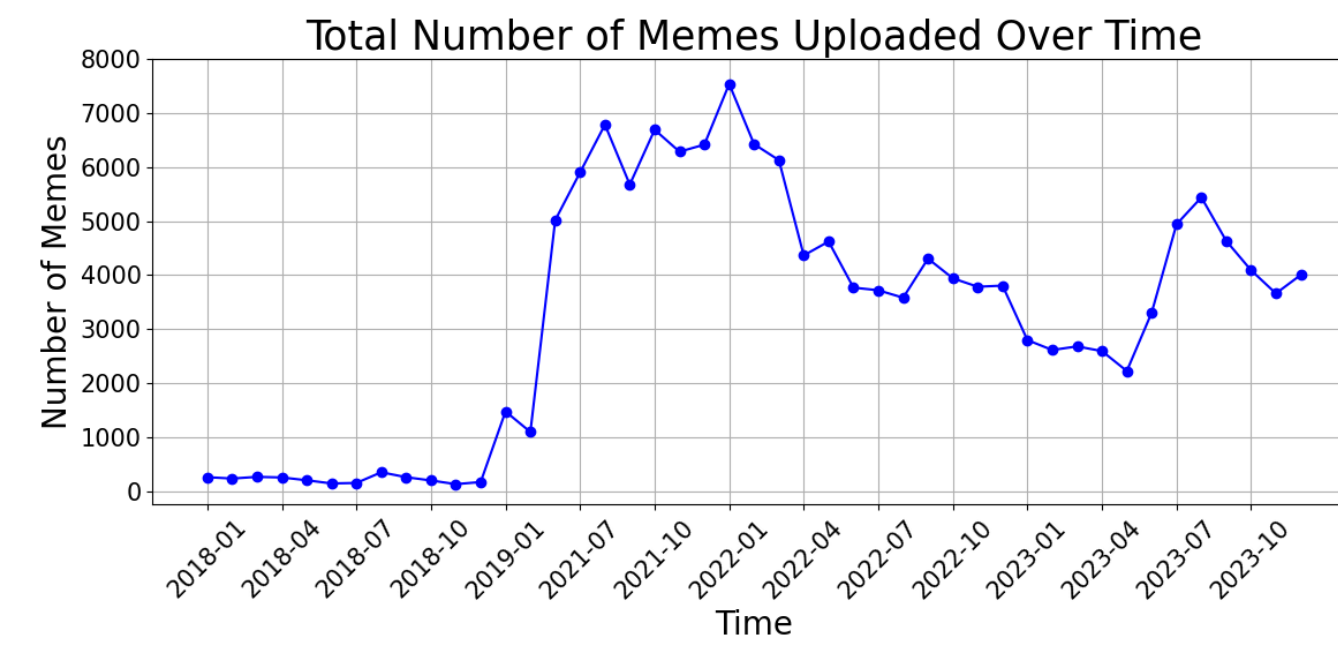
Schema (derived annotations) for MemeMatch v1.0.

Methodology



1. Data Collection and Preprocessing (Input Stage)

- 1.1. **ImgFlip**: 3,127 templates (Apr 2025) → **2,083 unique** after dedup; **153,792 template-labeled memes** (≈ 74 per template).
- 1.2. **Reddit**: Jan 2018–Dec 2023, up to **1,000 posts/day** → **899,522 images** with metadata; upvotes reflect the **Jan 2024 crawl window** (not lifetime totals).
- 1.3. **Deduplication**: Removed duplicates using **perceptual hashing (pHash)**.
- 1.4. **Final combined corpus**
 - **146,991 Reddit memes** (no template labels).
 - **153,792 ImgFlip memes** across **2,083 templates**.
 - Combined dataset is the input to the **Dual-Context Framework**.



2. Dual-Context Framework Stage

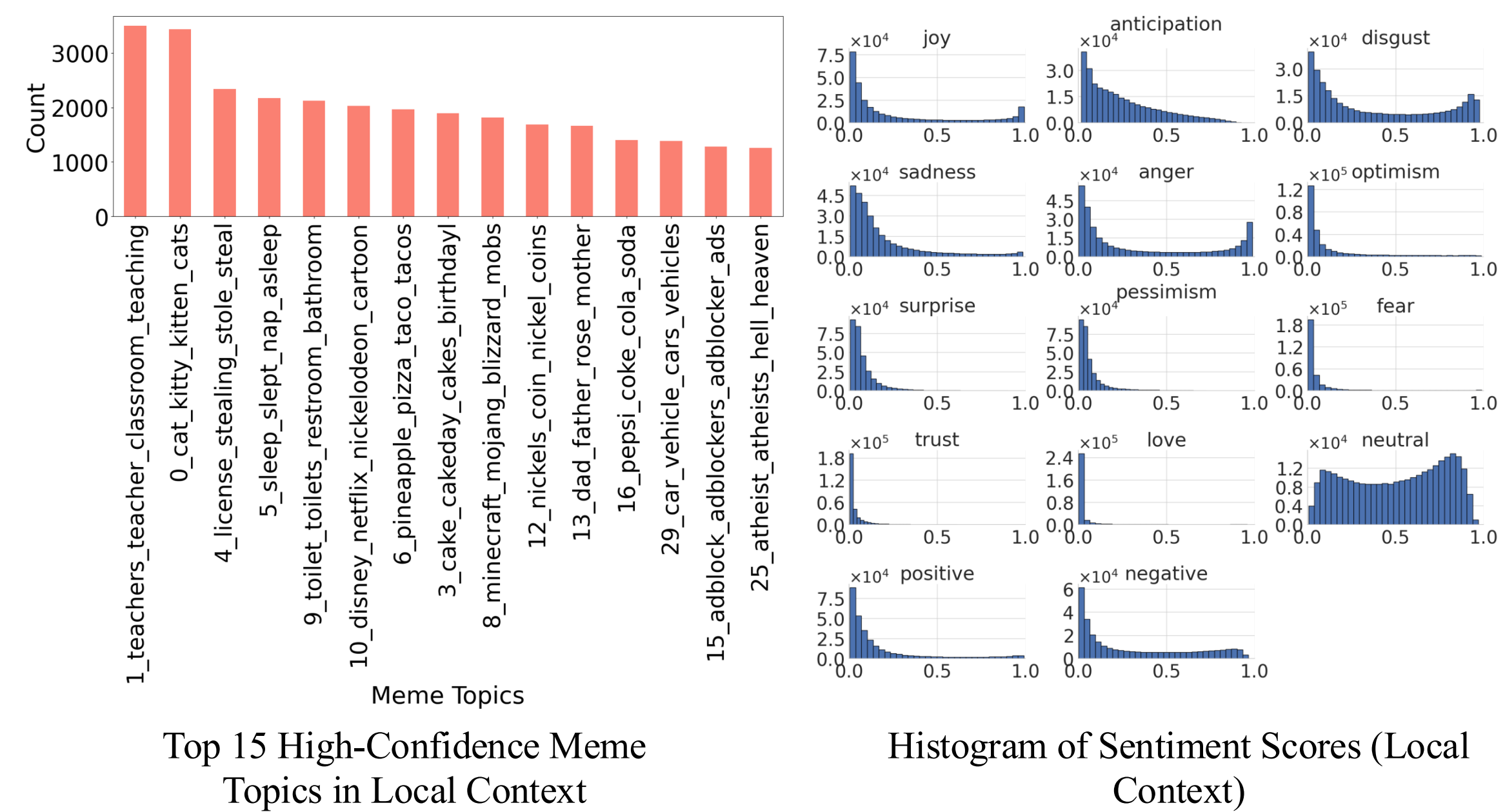
- 2.1. **Local Context Extraction (Message-Level)**
 - **EasyOCR** extracts overlay text; remove common template artifacts by filtering strings found in **2,083 templates** (e.g., watermarks/URLs), then append **Reddit title** → **local_context**.
- 2.2. **Global Context Extraction (Template-Level)**
 - **PaddleOCR** locates text to mask; **BLIP** captions the masked base image → **global_context** (for Reddit memes, “template” = base image after masking).
- 2.3. **Why Hybrid OCR?**
 - **EasyOCR** is strong for fast text extraction at scale, while **PaddleOCR** provides tighter boxes for masking before captioning.
- 2.4. **Output**: Each meme produces a paired representation: **local_context** + **global_context**, which are used in downstream annotation and retrieval (Annotations Stage).

3. Annotations Stage

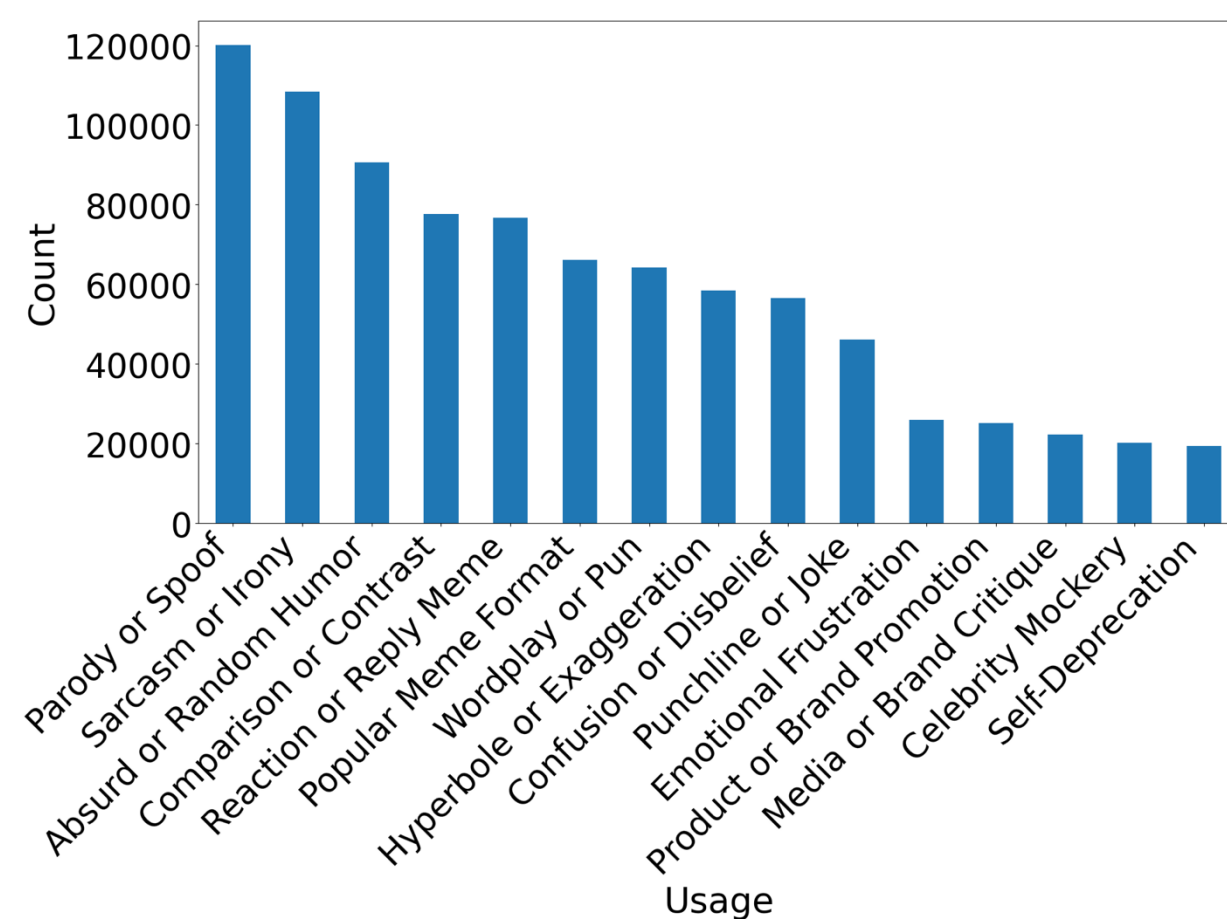
- 3.1. **Sentiment & Emotion (How the meme feels)**

We apply two **RoBERTa**-based models (CardiffNLP / TweetNLP) to **both local and global contexts**:

 - **Emotion model**: predicts **11 emotions** (*Anger, Anticipation, Disgust, Fear, Joy, Love, Optimism, Pessimism, Sadness, Surprise, Trust*).
 - **Sentiment model**: predicts **3 polarities** (*Positive, Neutral, Negative*).
- 3.2. **Usage Intent Labels (How the meme is used)**
 - We infer communicative intent using **zero-shot classification** (BART-MNLI) over **28 predefined usage labels** on concatenated **local + global** text; keep labels with **confidence ≥ 0.70** .
- 3.3. **Topic Modeling (What the meme is about)**
 - Cluster themes in both contexts: **300 local topics** and **200 global topics**; each meme gets a **topic ID + probability**, with low-confidence items marked as **outliers (-1)**.
- 3.4. **Auxiliary Feature**
 - **Text length**: character count of the **local context** (text_length), which complements higher-level annotations and supports downstream analysis.



Text	“3DS owners: OH NO! We'll have to pirate games now!” Wii owners:”
Emotion	
Anger	0.479241
Anticipation	0.199450
Disgust	0.647908
Fear	0.066868
Joy	0.032594
Love	0.002387
Optimism	0.016223
Pessimism	0.075464
Sadness	0.217513
Surprise	0.060239
Trust	0.005590
Negative	0.768447
Neutral	0.215292
Positive	0.016261



MemeMatch: Context-Aware Multimodal Meme Retrieval

Goal: given a **text query q** or **meme image I** , return a ranked list that matches:

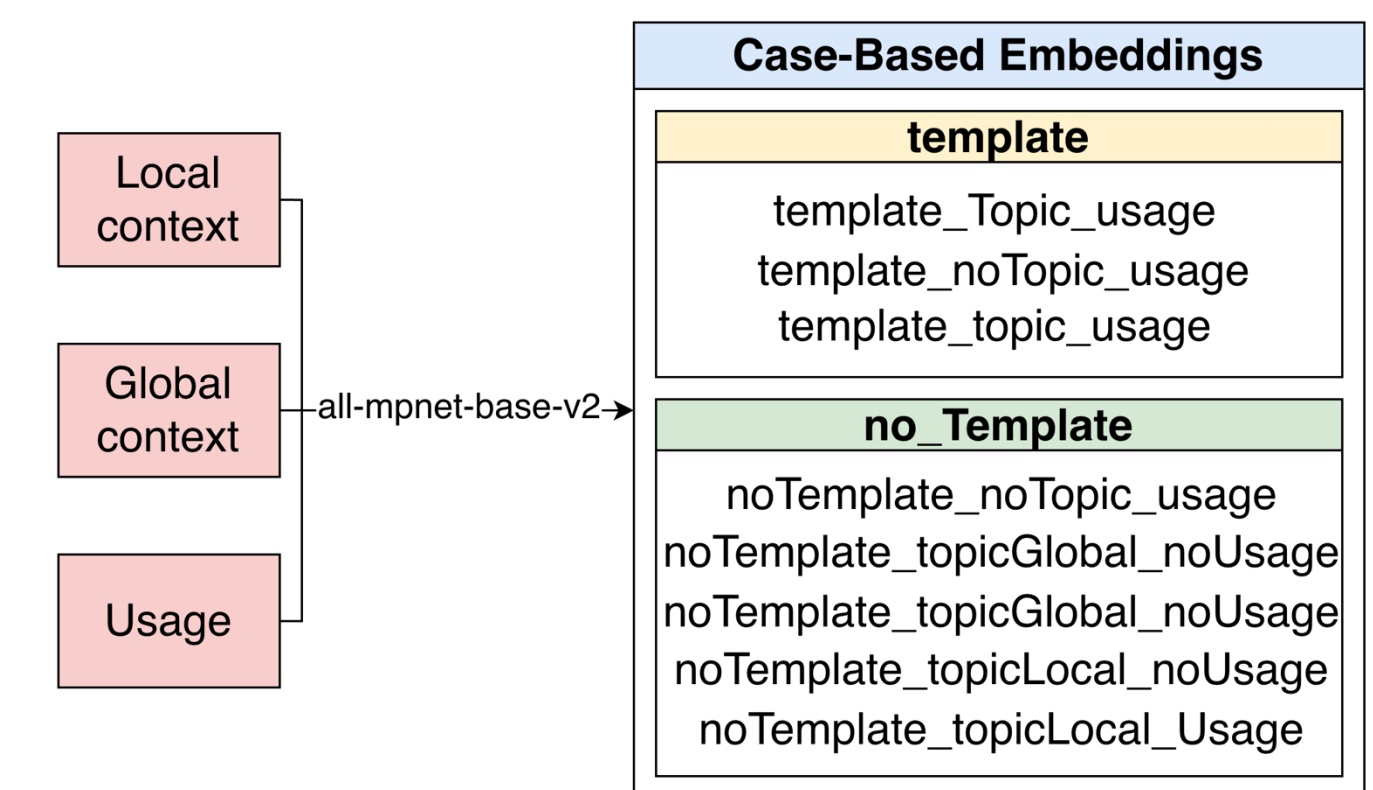
- **what the meme is about** (topic/meaning), and
- **how the user wants to use it** (intent, e.g., joke, complain, motivate).

1. Design Principles

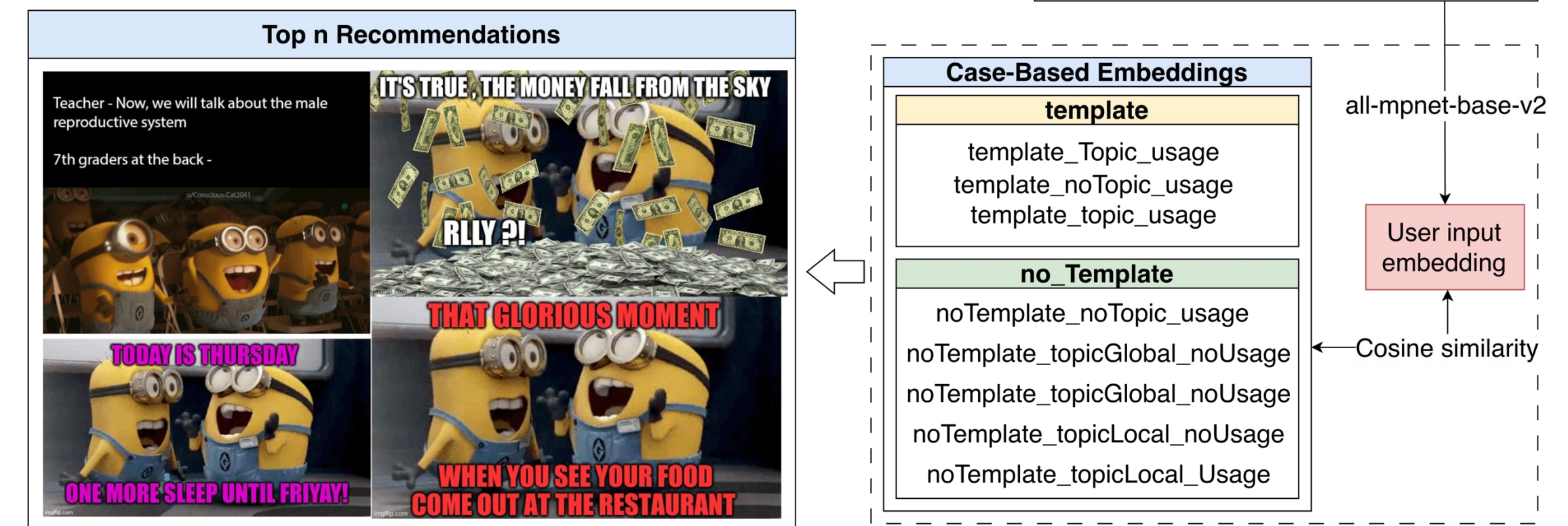
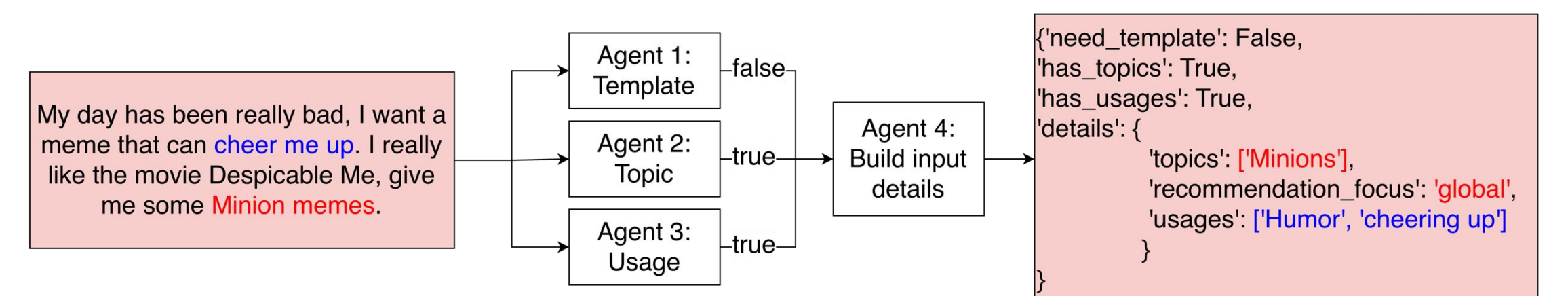
- **Context-aware**: uses both **Local** (overlay text + title) and **Global** (template caption) views.
- **Intent-aware**: incorporates **usage labels** (e.g., sarcasm, parody).
- **Efficient**: fast retrieval using **precomputed embeddings** + **cosine similarity**.
- **Responsible**: basic filtering and duplicate suppression.

2. Case-Based Embeddings (Fast Retrieval Backbone)

- Different queries contain different information (topic vs intent vs template) so MemeMatch precomputes multiple **embedding indexes (“cases”)** so it can retrieve efficiently for each query type.
- **Encoder**: SentenceTransformers all-mpnet-base-v2.
- Retrieval uses **cosine similarity** between the query embedding and the selected index.



3. Natural-Language Retrieval (Text Queries)



Users enter free-form text queries (e.g., “a funny meme about finals”); an LLM-based query parser extracts:

- **scope** (templates vs memes),
 - **topic** (e.g., Minions/exams),
 - **intent** (e.g., humor/motivation/complaint),
- then routes to the right embedding index and retrieves top-k results by similarity.
Fallback: If topic/intent is unclear, MemeMatch falls back to **tone matching** (e.g., “funny” → joyful memes) using sentiment scores.

3. Image-Query Retrieval (Upload a Meme)

Users can upload an image and retrieve similar memes in **two complementary modes**:

- **Global (template) mode**: mask text → caption underlying image → retrieve memes with similar template meaning.
- **Local (message) mode**: read overlay text → retrieve memes with similar text/message.

